

Chetna Pal

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EDUCATION

Integrated M.Tech in Computational and Data Science
VIT Bhopal University | CGPA: 8.32/10 | Expected May 2027

PROJECTS

- **SafeSphere – Community Safety Risk Modeling System**
 1. Designed a CNN-LSTM model leveraging geospatial and temporal data to assess commuter safety risk, achieving 87% accuracy in predicting high-risk zones during peak hours and providing actionable insights.
 2. Engineered an interactive, map-based risk visualization frontend using Folium and connected it to a modular FastAPI + Streamlit architecture; system now used by 15+ analysts.
 3. Improved risk prediction accuracy by analyzing location, time of day, and incident trends.
- **Netflix TV Shows Data Analysis | Python, Pandas, Matplotlib, Google Colab**
 1. Cleaned and preprocessed **1,000+ TV show records**, handling missing values and inconsistent data using **Python and Pandas**.
 2. Conducted **Exploratory Data Analysis (EDA)** to identify trends in release years, genres, and audience ratings.
 3. Built **5+ data visualizations** using **Matplotlib** to analyze rating distribution and yearly content growth.
 4. Spearheaded the transformation of raw, unformatted data into a structured, analysis-ready format, reducing manual processing time by 75% for downstream applications.
 5. Published complete project workflow on GitHub, demonstrating version control and reproducible analysis.

SKILLS

Technical Skills

- Programming Languages: Python, C++, SQL
- Machine Learning: CNN, KNN, Model Evaluation
- Data Analysis: Pandas, NumPy
- Data Visualization: PowerBI, Tableau
- Databases: MySQL, Oracle
- Cloud Technologies: AWS (EC2, S3, IAM – basics)
- Core Concepts: Data Structures, Algorithms, DBMS, OS, Computer Networks
- Tools: Jupyter Notebook, VS Code
- Additional Skills: Data Collection, Data Cleaning and Preparation, Inventory Management, EDA

RELEVANT COURSEWORK

- Machine Learning
- Data Structures and Algorithms
- Database Management Systems
- Operating Systems
- Computer Networks

CERTIFICATIONS

- Simulink Onramp by Mathworks
- Matlab by Mathworks
- Introducing Generative AI with AWS by Udacity
- Applied Machine Learning in Python by University of Michigan
- MERN Full stack with AI Internship Program by Ethnus
- The Ultimate Job Ready Data Science Course by Code With Harry

EXTRACURRICULAR

- Active member of the **Data Science Club**; volunteered in technical events and hosted sessions to engage participants.
- Led and organized **community teaching initiatives** for children of construction workers, focusing on basic literacy and

numeracy.

- Participated in multiple **musical performances and talent hunts**, demonstrating teamwork, creativity, and stage presence.
- Lead dancer in official **VIT Bhopal ad shoot** .

SUMMARY

I am a M.Tech student specializing in Computational and Data Science with hands-on experience in machine learning, data analysis, and backend development. Proficient in Python, FastAPI, AWS, SQL, and data visualization tools, I have developed real-world AI-driven projects that demonstrate my technical skills and problem-solving abilities. I'm committed to leveraging data for actionable insights.

